

David León Salas

Python Backend Developer

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About me

Python Backend Developer with hands-on experience building RESTful APIs using FastAPI and Flask, applying clean architecture across 3 internships. Familiar with Docker, CI/CD, and Azure deployments. Completing a Computer Systems Engineering degree (90/100 avg, graduating May 2026). Fluent in English (B2) and native Spanish speaker.

Education

TecNM - Campus Instituto Tecnológico de Veracruz, Bachelor's in Computer Systems Engineering – Veracruz, Mexico Aug 2021 – May 2026

- GPA: 90/100.
- Graduate of the Samsung Innovation Campus (AI & Leadership program).
- Member of the Competitive Programming Club (CPC).

Experience

Data Analyst Intern, TenarisTamsa – Veracruz, Mexico Feb 2026 – present

- Automated daily ingestion from 5+ data sources (CSV, XML, Access, Excel, PostgreSQL, SQL Server) using Python and batch scripts, saving ~1 hr/day of manual processing across recurring workflows.
- Developed multiple web scrapers using Python and Selenium to automate data downloads from internal systems, packaged as standalone executables (.exe via PyInstaller) and deployed on a virtual machine for scheduled, dependency-free runs.
- Contributed to the development of a PowerApps application for full tube defect cycle management — centralizing workflows previously handled via Excel and email into a single system, using PowerFX for backend logic.
- Contributed to the development of a Power BI dashboard for CTT (Carga al Mil de Acería) steel mill load tracking, integrating SQL queries and Python scripts for data calculations and preparation.
- Invited to present Python automation work in knowledge-sharing sessions for fellow interns.

Python Backend Developer Intern, NeoDigital – Veracruz, Mexico Mar 2025 – Oct 2025

- Built a modular RESTful API (FastAPI + MongoDB) to manage and query official SAT catalogs for internal tax compliance workflows.
- Implemented endpoints to upload Excel files, convert them to structured JSON, and store them in MongoDB using an async driver (Motor).
- Enforced data integrity using Pydantic models and layered architecture (routers / services / repositories / utils).
- Wrote a bulk conversion script to process SAT Excel catalogs into JSON, reducing catalog onboarding from a manual multi-hour task to an automated process.

Python Backend Developer — Research Internship, Instituto Tecnológico de Veracruz – Veracruz, Mexico Dec 2024 – Dec 2025

- Developed REST API endpoints with Flask, integrating Ollama open-source LLMs with streaming responses for a conversational AI research tool.
- Applied clean architecture (controllers / services / repositories / utils) to keep the codebase maintainable and testable.
- Deployed the application on Microsoft Azure App Services using Docker, with reproducible builds through a CI/CD pipeline.

Projects

Personal Portfolio — dvleu.dev – Veracruz, Mexico

Jan 2026 – present

Personal portfolio website built from scratch with React, TailwindCSS, and Vite — deployed with a custom domain.

- Designed and developed the full site from scratch without AI code-generation tools, covering UI, layout, and responsiveness.
- Built with React and TailwindCSS, bundled with Vite, and deployed to production at dvleu.dev.
- Source code available at github.com/DvLeu/Portafolio.

HealthMed Chat — Medical Diagnosis API – Veracruz, Mexico

Jan 2024 – May 2024

Academic backend project — a Python API that infers probable diagnoses from patient symptoms using rule-based filtering and KNN scoring.

- Designed an API receiving patient attributes (age, gender, BMI, symptoms) and returning a ranked Top-5 diagnosis list with confidence scores.
- Built a two-stage engine — symptom/gender filtering followed by KNN + classical scoring to rank results.
- Added session logging to capture inputs, outputs, and metrics for analysis.

Lnap – Veracruz, Mexico

Dec 2024 – Dec 2025

Research project developing educational programming tools under an institutional contract.

- Built an interactive visualization tool to explain programming structures across multiple languages, used in academic sessions.
- Developed a pseudocode-to-Python translator to help students transition to formal syntax.

RustParser – Veracruz, Mexico

Jan 2024 – May 2024

Academic compiler design project — lexical analyzer and parser for Rust syntax using Java and JavaCC.

- Defined tokenization and grammar rules for Rust constructs (variables, functions, control flow, expressions).
- Validated structural correctness of Rust source files via custom JavaCC grammar, supporting comments and arithmetic/logical expressions.

Skills

Specialty: Python Backend: FastAPI, Flask, Django, REST API design, async Python, Pydantic, clean architecture

Databases: MongoDB, PostgreSQL, Microsoft SQL Server, MySQL, SQLite

DevOps & Cloud: Docker, CI/CD, Linux, Microsoft Azure (App Services), Git, GitHub

Data & Automation: Power BI, DAX, Power Query, PowerApps, web scraping, Excel automation

Additional Languages: Java (Spring), JavaScript, TypeScript, SQL, Shell Scripting

Concepts: Layered/clean architecture, OOP, REST principles, agile/scrum

Certifications

Samsung Innovation Campus (2025): Artificial Intelligence & Leadership — Samsung Electronics & Universidad de Monterrey

IBM AI & Python Specialization (2025): Python for Data Science, AI Development with Flask, Introduction to AI, Hands-on Linux & Git

Cisco Data Analytics Essentials (2025): Data analysis fundamentals, visualization, and business insights

Cisco Python Essentials 1 & 2 (2025): Python programming fundamentals and intermediate concepts

6th Autumn School on Distributed Systems (2025): Distributed systems and next-generation network concepts

Cisco Introduction to Cybersecurity (2025): Core cybersecurity principles and threat awareness